Engineering Methodology

Ablation Study: ReGainX Model vs. Baseline Model

Training Performance

Figure 33: PPO vs baseline- Created by Researcher

★ Base PPO

★ RGX PPO

★ **87.74% accuracy** (↓9.35% vs. ReGainX)

★ **37.6%** ↓training performance

★ Elementary approach
- **Ablation Study**
- Default hyper params
- Low exploration, **lr**, & rollouts(<100)

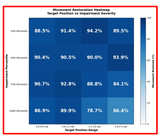

Figure 34: Baseline Matrix - CBR

Temporal Difference Error

★ **Critic's prediction error** for future rewards

★ Reduces TDE by **61.4%** → stable predictions
- STDEV = %0.55
- **Feature ext. + tuned hyper params**

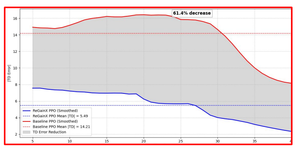

Figure 35: ReGainX vs. Baseline TD Error - Created by Researcher

★ Base PPO

★ RGX PPO

TD Err. Equation

$$T = R_t + \gamma V(s_{t+1}) - V(s_t)$$

R_t = reward at timestep(t)

$V(s_{t+1})$ = future reward *after* R_t

$V(s_t)$ = future reward *before* R_t

Conclusion

★ This research shows that exoskeletons can be used to **(a)** assess movement and impairment severity, **(b)** mitigate impairment in real-time, and **(c)** monitor PD progression.

Engineering Criteria

★ **Natural, effective real-time functionality**: 100ms assistance intervals & high accuracy

★ **Adaptive across all users:** Patient-specific processing & evaluated across varying conditions

★ **Safe, non-invasive:** Non-invasive sensors & low-risk product

★ **User-friendly:** Intuitive control, easy-to-use velcro straps, user personalization through app

Key Outcomes

★ **Overcomes existing limitations**: low-cost, state-of-the-art exos
- No fixed assistance strategy → adaptable

★ Custom feature extractor to process EMG & IMU streams to guide **prediction of user movement & impairment severity**
- Utilizes two different tasks to mitigate and track motor impairments

★ **Proof of concept:** RL-driven, patient-specific, daily-use tech
- Lack of heavy computation, patient data, & resources
- **Scientific value**

★ Unique application of RLHF → patient is "expert"

97.8%
Elbow Movement Restoration

95.5%
L-L Movement Restoration

96.9%
Correlation with PD severity

Impact

★ **ReGainX enhances quality of life** for PD patients, primarily those lacking access to medication
- **Reduced muscular fatigue, delay, & pain** → improves **independence** & daily functioning
- Reduces morbidity and **mental effects of PD**

★ PD severity monitoring → patients & clinicians can monitor progression through app
- **Eliminates need for costly, time-consuming clinical examination**

★ **Accessible** & practical real-world-use
- **Low-cost**, modular exoskeletons targeted at PD symptom sites.

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